ZCAS University

ALL FIRST YEAR PROGRAMMES - SOCTAS

CCS1412 – FUNDAMENTALS OF DIGITAL LOGIC DESIGN

FINAL EXAMINATION

2ND DECEMBER 2024

14:00 HRS-17:00HRS

TIME ALLOWED: WRITING – THREE HOURS

READING – 5 MINUTES

INSTRUCTIONS:

1. Section A: this question is **compulsory** and must be attempted.
2. Sections B: Answer **THREE (3)** questions from this section.
3. This examination paper carries a total of **100 marks**.
4. **NO** calculators allowed.
5. Candidates must **not turn this page** until the invigilator tells them to do so.

SECTION A: Question 1 is compulsory and must be attempted.

Question 1

a) A finite state machine (FSM), otherwise sequence detector has input A and output Y. The output Y = 1 when the sequence 0101 has been inputted through input A on the last 4 consecutive rising clock edges (or ticks). Otherwise, Y = 0

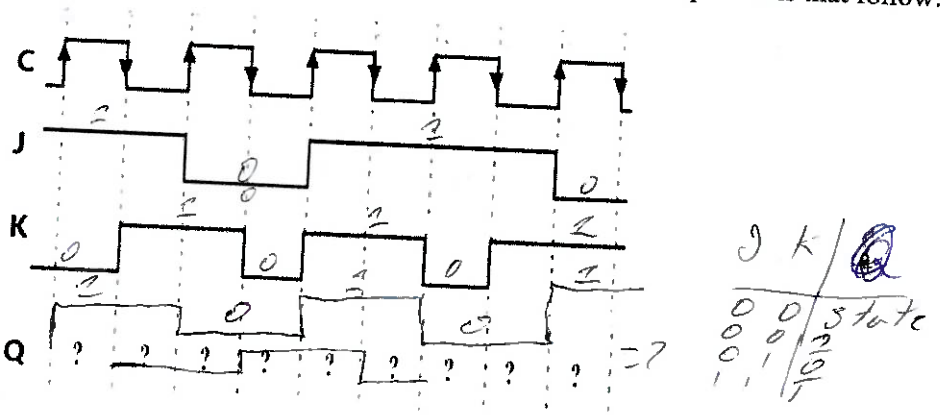
i. Draw state transition diagram that corresponds to the above requirements. (The first state should be S_0). **[10 Marks]**

ii. Fill in the table below to complete the various state assignments possible for such as system:

State	One-hot	One-cold	Binary	Gray code
S_0	*Null*	*Null*	*Null*	*Null*
S_1	1000			
S_2	0100			
S_3	0010			
S_4	0001			

[16 Marks]

b) Flip-flops are the basic building blocks of memory registers. Consider the waveform and data inputs of a JK flip-flop shown in Figure 1 below and answer the questions that follow.



i. Given that the above inputs C, J, and K are inputs to the master flip-flop of a JK M-S flip-flop, redraw the above timing diagram and generate the output Q representing the output on the **slave**. **[5 Marks]**

ii. Draw a JK flip and explain its operation using a truth table. **[5 Marks]**

iii. Describe the specific issue that leads to a race-around condition in the JK flip-flop when both inputs are set to high. **[4 Marks]**

Total: (40 Marks)

SECTION B: Attempt any THREE questions in this section.

Question 2

Truth tables are an indispensable tool in the design and analysis of digital systems. They facilitate the specification, design, verification, testing, and optimization of digital circuits and systems, ensuring that they meet their intended functionality and performance requirements. "Don't Care" conditions are conditions where the output value is not critical or doesn't matter, allowing for additional simplifications in the logic design. Consider the minterm list $\sum m(0,1,4) + \sum m(5,6)$ of a particular system shown below and answer the questions that follow.

i. Draw the corresponding truth table that represents the minterm list equation above. **[5 Marks]**

ii. Draw the Karnaugh map of the system and use it to obtain the minimised and optimal Boolean expression. Draw the resultant circuit using basic gates. **[10 Marks]**

iii. Consider the following column headings of a k-map: 00, 01, 11, 10. Explain why the sequence changes from 01 to 11 and not to 10. **[5 Marks]**

(20 Marks)

Question 3

Answer the following questions justifying your answer **[2 Marks each]**. ***NOTE: YOU GET A MARK FOR CORRECT ANSWER WITHOUT JUSTIFICATION, YOU GET NEGATIVE (-1) FOR WRONG ANSWER AND ZERO (0) FOR "I don't Know":***

i. Which of the following is a universal gate?
A. OR B. NOR C. XNOR D. I don't Know.

ii. Which of the following is a basic building block of a register?
A. D flipflop B. JK Logic Gate C. Transistor D. I don't Know.

iii. Which of the following number systems can be represented by 4 bits?
A. Binary B. Octal C. Hexadecimal D. I don't Know.

iv. What are the terms in SOP called?
A. Minterms B. Maxterms C. Boolean D. I don't know.

v. Which of the following is a basic building block of a FET?
A. Universal gates B. Basic gates C. Derived gates D. Semi-conductor E. I don't Know.

vi. Which of the following sequential elements resolves the forbidden input combination in SR flip-flops?
A. Master-salve B. JK C. T D. D E. I don't Know.

vii. The number 1024 when converted to binary is _____?

A. 10101101 B. 11111111 C. 100000000 D. 1111111 E. None of the above.

viii. When implementing binary adders, what is the sum of 1 + 1 + 1 + 1 = ?.

A. 0 B. 1 C. 10. D. 100 E. I don't Know.

ix. What do JK latches suffer from?

A. Race condition B. Race around C. Toggle D. I don't Know.

x. CMOS logic always has_____?

A. p-FET B. n-FET C. A and B D. Transistor. E. I don't Know.

(20 marks)

Question 4

Number systems play a crucial role in digital logic design, providing a foundation for representing and manipulating information in digital circuits of computing systems. The importance of number systems in digital logic design can be understood through several key aspects. Consider and answer the following questions:

(a) The MAC address of a WiFi network interface card is 00-1E-68-3F-73-E6. Convert this address to binary. **[5 Marks]**

(b) One of the current dongles (modem) that some mobile network provides has a maximum internet speed of 10.24 Mbps. How many Kilobytes of data can you download in half an hour using such an internet connection? **[5 marks]**

(c) Convert the decimal number 9.625 to binary. **[5 Marks]**

(d) Minimise the following Boolean function and draw the corresponding logic circuit.

$f(w, x, y, z) = (w \cdot \bar{x} \cdot (y + x \cdot z) + \bar{w} \cdot \bar{x}) \cdot y$ **[5 Marks]**

$= \bar{x} \cdot x(y + z) + (w \cdot \bar{w}) \cdot y$

(20 Marks)

Question 5

State whether the following is **True** or **False.** A correct answer earns you a mark and a correct justification earns you an additional mark. **[2 Marks each]:**

a) If you combine the inputs of a NOR gate into one, you get a NOT gate.

b) In CMOS logic, the n-FET are to be connect to POWER.

c) Computers use hexadecimal numbers to represent MAC addresses.

d) When you AND a variable with itself, the result is always a 1.

e) When you OR a variable with itself, the result is always a 1.

f) When you connect the two inputs of a JK flip-flop and add a NOT gate to the K input, the resultant flip-flop is a D flip flop.

g) Propagation delay is always longer than or equal to the contamination delay.

1024 2 512 2 256 2 128 2 64 2 32 2 16 2 8 2 4 2 2

1 0 0 0 0 0 0 0 0 0 0

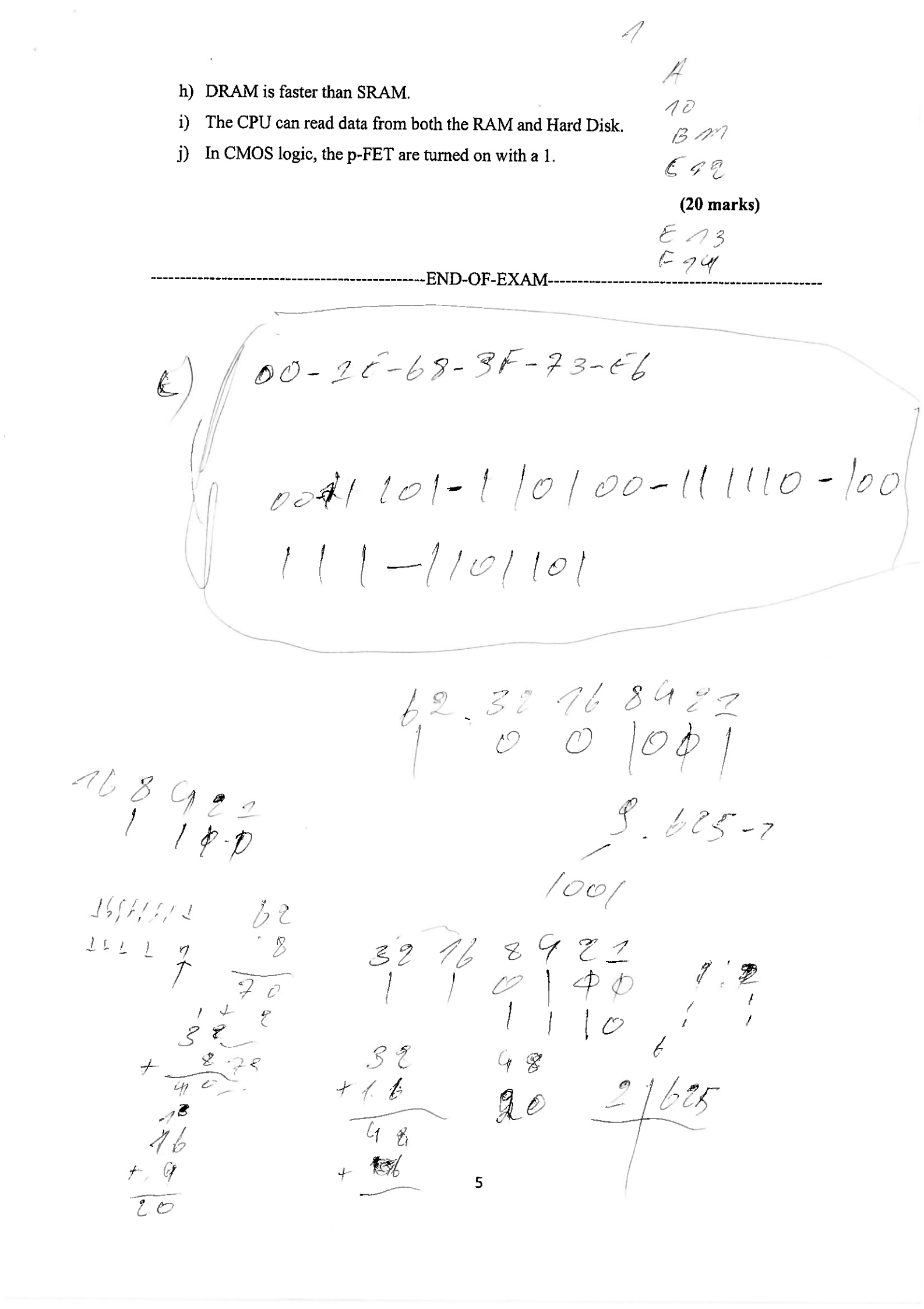
h) DRAM is faster than SRAM.

i) The CPU can read data from both the RAM and Hard Disk.

j) In CMOS logic, the p-FET are turned on with a 1.

(20 marks)

----------END-OF-EXAM----------



A	B	C	Q
0	0	0	1
0	0	1	0
0	1	1	
0	1	0	
1	0	0	
1	0	1	
1	1	1	